University of Padua

Department of Mathematics "Tullio Levi-Civita"

Master Degree in Computer Science

GeoMedia: A Social Media for Location-Based and Time-Sensitive Content Sharing

Master's Thesis

Supervisor

Prof. Claudio Enrico Palazzi

Co-supervisor

Dott. Lorenzo Perinello

Candidate

Alberto Morini

Student ID
2107783

Academic Year 2025-2026

Summary

Over the past few decades, the use of mobile devices has increased drastically, almost replacing desktop software and web browsers. This shift has been mainly driven by the widespread adoption of smartphones not just for phone capabilities but also for the availability of integrated sensors such as GPS, gyroscopes, and Bluetooth, which allow richer and more context-aware user experiences. By leveraging these technologies, Online Social Network (OSN) platforms have been able to provide more advanced and sophisticated features; consequently collecting larger amounts of metadata, interactions and daily engagement.

This thesis presents “GeoMedia”, an open-source framework designed for sharing multimedia content — such as text, images, audio, and files — anchored to real-world geographic locations. By leveraging GPS data available on mobile devices, the application provides an interactive map through which users can explore and publish digital content, creating a seamless integration between the digital and physical worlds. Moreover, the framework offers several features and covers various use cases, including restricting content visibility within a customizable distance range, enabling time-sensitive content or applying user-based restrictions for content creation and visualization. It also supports sequential visibility between published posts thus satisfying various usage scenarios and discovery features among published data.

Furthermore, it analyzes the motivations behind GeoMedia, its critical aspects and design considerations, describes its system architecture and implementation through a comparison of different technological paradigms. Finally, it explores potential applications and outlines possible future developments of the framework.

A previous version of this framework has been presented at GoodIT 2025: International Conference on Information Technology for Social Good (September 3-5, 2025 - Antwerp, Belgium) [1]

Contents

List of Figures

List of Tables

List of Codes

Chapter 1

Introduction

1.1 Overview

Technology has always influenced society, from the first introduction of the Personal Computer to the Internet, changing the world year after year; from personal life to workspaces, people are surrounded by software on their computers and mobile devices. Over the last decades, the investments in the development of technological innovations, both in hardware and software, as increased exponentially, shaping economical and social impact by integrating the latest features into everyday life. Smartphones and tablets have been the core engine that completely changed daily habits and digital usage; their wide adoption has reached almost three quarters of global population [54, 49], making users abandon traditional computers, expecially for social media [48] and instant messaging platform [51], this trend bring a change on the design and development of applications into a 'mobile first' logic.

Compared to computers, mobile devices offer distinct functionalities and strengths that have facilitated their widespread adoption. These can be categorized according to the fundamental logic underlying any technology:

1. **Hardware**: the increase in computational performance[32], battery life[6], fast connectivity, and the integration of several built-in sensors[31] such as gyroscopes, GPS, NFC, light sensors, cameras, and vibration motors. These enable applications to interact with different fields such as health[4], education[37], and, through motion-sensors or cameras, an interaction with physical world, creating a more immersive user experience.

2. **Software**: the development and presence of mobile applications has played a crucial role. In fact, platforms such as Windows Phone were eventually abandoned and discontinued due to the lack of applications [5, 56] compared to operating systems like 'iOS'[14] and 'Android'[17].

In this way, smartphones apps have became a key integration with daily life eand physical world, for example, when planning a trip or just driving, smartphones, integrated with cars assists users via app like "Google Maps"[19] or "Waze"[12], providing not just directions but also real-time traffic updates, read events or information about available parking. This makes the surrounding environment enriched with digital details. Another example is 'Google Lens'[18], which can identify objects from a photograph, or "Shazam"[44] that through devices‘ microphone can recognize a playing song. Even in workplaces or public areas, technology integrations can be present, from the resturant that provide a digital menu by a tablet, to customers checking the nutrition macros into a grocery store by scanning the barcode on the pack, to museums by scanning a QR Code enabling audio-guided tours or unlock Augmented Reality functionalities.

1.2 Online Social networks

Alongside the widespread adoption of smartphones and mobile devices, online social networks, have emerged as one of the most influential and important sectors of the technology market. Today, social media platforms rank among the firsts most used applications worldwide [39, 13], while the companies that provide them become some of the most bigger actors in global financial markets [40, 28, 24].

Online Social Networks (OSN) are the digitalized form of a Social Network (typically represented as a graph of people where links correspond to relationships) through the Internet[41]. For instance, Facebook, one of the largest platforms, represents a graph of friendships, where nodes (users) are connected via undirected edges, assuming that friendships are mutual. There are also different types of OSN, such as Instagram or Twitter, where the links between users are directed, implementing a following model.

These services provide a variety of functionalities, first of all instant messaging capabilities, both one-to-one and one-to-many, and, in addition, the ability to share a wide range of multimedia content, including images, videos, audio recordings, and other types of documents.This facility and advantages compared to other communication forms, like SMS, has led to their integration across diverse social, professional, and public contexts. The particularity of sharing multimedia content, commonly referred to as a post, has coined the term Social Media, where the published content represent the node of interaction between users through comments or reactios (e.g., ‘like’ or ‘unlike’), thus to replicate human behavior in dialogues and social interactions.

Despite the significant benefits offered by social media services, including enhanced connectivity, community creation, and information sharing, their adoption has also introduced various social and ethical challenges. Concerns regarding user privacy are particularly relevant, as it is often not respected by the service providers themselves, who collect personal information and interests in order to monetize them or enable surveillance practices[29]. Other serious issues include the dissemination of misinformation and fake news, nowadays further amplified by Artificial Intelligence-generated content, commonly known as ‘deepfakes’ [30]. Such content can influence public opinion and social behavior, not always in a positive way [53, 33].

Online Social Networks have also intensified common societal problems through phenomena such as ‘echo chambers’[3], where users are repeatedly exposed to information and content that aligns with their existing beliefs and preferences. This continuous reinforcement can strengthen existing views, reduce exposure to alternative perspectives, and, in some cases, contribute to ideological radicalization. In practice, examples can be observed in so-called ‘incel’ communities [34] and among supporters of extreme political movements.

To address these issues, social media platforms have adopted various moderation and safety mechanisms, including automated content analysis systems, community guidelines and reporting tools. While these measures contribute reducing the spread of harmful content, they are not always fully effective, due to the scale and complexity of emerging innovations, such as ‘deepfake’ [2]. Consequently user reporting and human moderation aside of automated tools, continue to play a crucial role in identifying, evaluating, and removing content that violates policies or legal regulations[9].

Chapter 2

State of Art

GeoMedia positions itself among existing online social media platforms, by offering functionality centered on the sharing of geolocated content linked to specific latitude and longitude coordinates. Content is visible only within a configurable distance radius and can be more restricted with time-sensitive parameters or user-based policy. Thus to create a wide range of real-world applications, including scavenger hunts, interactive city guides, and any other location-based experiences.

The platform enables users to discover and interact with the physical environment through a map enriched by content catagorized into different topics collections, contributed by other users, who can upload pictures or other multimedia resources associated with specific locations.

Although several other social media platforms utilize the GPS capabilities of mobile devices to enhance user experience and provide location-aware features, none fully realize the underlying concept of GeoMedia or offer the same degree of flexibility for creating diverse application scenarios and exploration of uploaded content.

2.1 Foursquare Swarm

Foursquare Swarm is the spin-off of a formerly app, "Foursquare City Guide", which provide personalized recommendations of nearby places to the users' current location based on its interests or previous browsing history and check-ins (locations previously visited by the user).

Foursquare was discontinued in 2024 and its functionalities have been integrated into Swarm, which in addition allows users to share their position with relatives, maintaining a personal lifelog of visited places, which can be shared with others.

For each visited location, users may leave reviews or upload multimedia content; however, these contributions are strictly tied to physical place, such as stores, cafés, restaurants or other kind of attractions as shown in Figure 2.1. Users cannot publish content at arbitrary geographic coordinates, nor can restrict post visibility, as all content is publicly accessible. The application further encourages the usage through a gamification system, in which users earn badges for visiting locations and uploading content. These mechanisms are broadly comparable to those implemented in other platforms such as Google Maps (see Section 2.4).

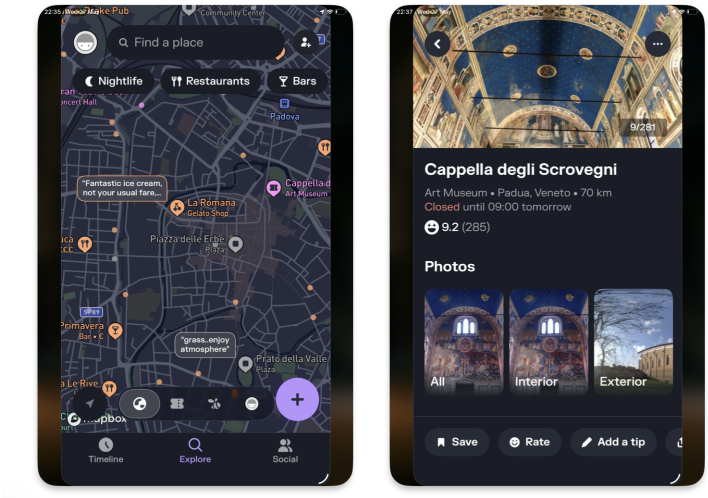

Figure 2.1: Foursquare Swarm show case

2.1.1 Dodgeball and Google Latitude

Dodgeball was a service that simply allowed users to share their current locations with friends through a map, and served as a precursor to Foursquare City Guide and Foursquare Swarm same functionality. Later was acquired by Google and rebranded as 'Google Latitude' [22] discontinued in 2013 and integrated into another discontinued social media by Google (Google+ [20]) and finally migrated into Google Maps 2.4 in 2018.

2.2 Snapchat

Snapchat [27] has been one of the most used social media platforms worldwide [26], mainly thanks to its instant messaging service and multimedia functionality, such as time-based content, which makes uploaded content visible only for 24 hours or visible only once to a user. This encourages users to check it daily in order not to miss anything and to keep up with relatives' lives. As a negative effect, these features have fueled some social impacts such as FoMO (Fear of Missing Out) [10, 43], leading users to become increasingly attached to the social platform and also creating self-comparison with content posted by other users. This problem is not strictly related to Snapchat, but is very common in every self-centric social media platform. Other concerns are raised by privacy and safety dued to the possibility to share current location into a geographic map [35], as done by Swarm and others (Section 2.1), Snapchat also enrich the map by showing geographic closer content to user, grouped into a unified cluster of information, as illustrated in Figure 2.2.

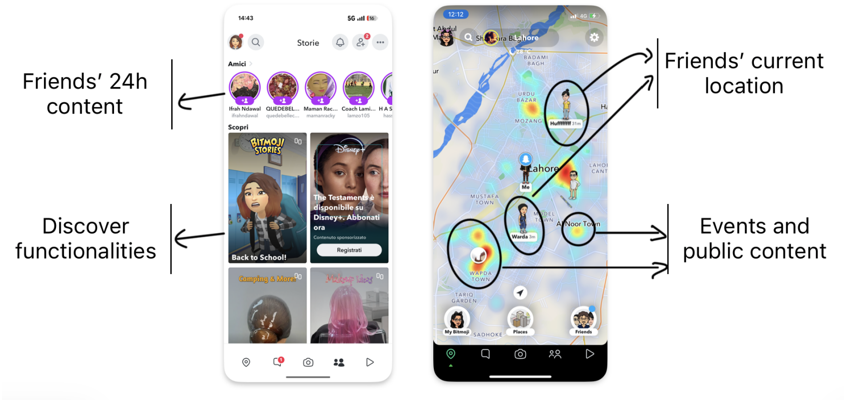

Figure 2.2: Snapchat functionalities

Moreover, as the Swarm app does 2.1, it allows users can log visited locations, including public places such as bars, stores, and museums. This enables the association of user-generated content with specific points of interest (POI) and supports the aggregation of geographically and contextually related content to represent events, such as concerts or other real-word gatherings, as illustrated in Figure 2.3

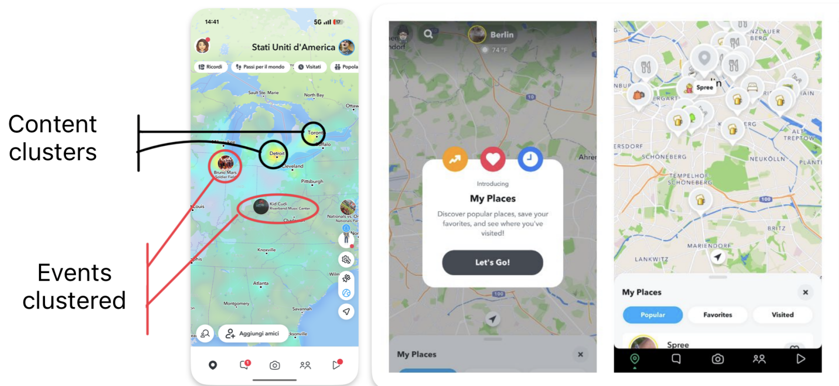

Figure 2.3: Snapchat 'my place' functionality

Snapchat represents a notable example of the adoption of the "mobile-first" paradigm, expecially in social media applications. The platform leverages not only GPS module but also a variety of mobile device sensors, including the camera for augmented reality (AR) experiences and the microphone to records voice-based content. By integrating these hardware features into its core functionality, the app delivers a highly immersive and personalized user experience.

2.3 Instagram

Instagram, owned by Meta Platforms, is one of the biggest and most used social media of the latest decades. Through the introduction of numerous features and continuous innovations, it has attracted more than two billion montly active users worldwide [42, 50].

The platform was initially released in October 2010 exclusively for iOS devices. In fact, at the time, the uploaded content was framed in a square (1:1) aspect ratio with a resolution of 640 pixels, reflecting the display characteristics of iPhones at the time. Later acquired by the formerly named Facebook Inc. (now Meta Platforms Inc.), the application was also deployed for Android systems in April 2012. Then, the original aspect-ratio limitations were removed, and several additional functionalities were introduced, including direct messaging and support for multiple images or videos within a single post.

A major development was the introduction of the "Stories" feature, which provide the same mechanism created by Snapchat 2.2, that allows users to publish content that remains visible for only 24 hours after publication. The addition of this ability contributed significantly to Instagram surpassing Snapchat in user engagement metrics by 2019 [52].

To enrich the media sharing, Instagram provides a variety of creative tools, including photographic filters, image-editing capabilities, and even a live-streaming functionality. The platform also popularized the use of hashtags as a content categorization mechanism, enabling users to discover media and connect with communities centered around shared interests and topics.

Instagram to encourage content discovery adopt a sophisticated recommendation systems that leverage contextual information, including geographic location derived from GPS or cellular network data. Such metadata$_g$ contributes to the generation of personalized recommendations, allowing users to discover nearby profiles, locations, and content that align with their interests [25].

Despite its popularity, Instagram has been the subject of significant criticism. Concerns have been raised regarding user privacy, large-scale behavioral profiling, and the potential influence of recommendation algorithms on user behavior. Additional controversies involve the platform's impact on adolescent mental health, content moderation practices, and the dissemination of illegal or inappropriate content.

2.3.1 Instagram Map

Despite its numerous functionalities, Instagram introduced in 2022 a map-based content discovery feature[7]. By anchoring content to existing geographic entities such as cities or others POIPoint of Interest (POI), users can explore both nearby or remote content, as illustrated in Figure 2.4.

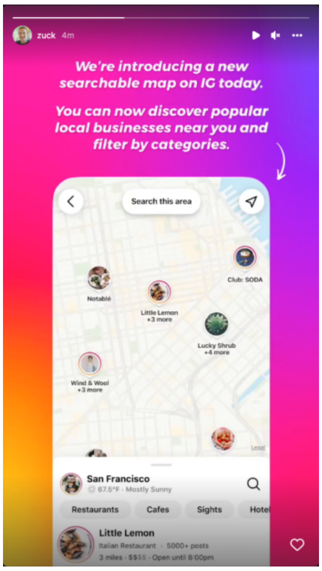

Figure 2.4: Instagram map discovery

In 2026, this functionality underwent a redesign: content is no longer presented as spatial clusters on the map (as implemented in Snapchat, Figure 2.3), instead is represented in a grid-based layout when a specific place is selected via the searchbar of the “Explore” section, as illustrated in Figure 2.5.
In addition, Instagram provides functionality for sharing real-time location with users connections (followers), similarly to other social media platforms discussed in Sections 2.2 and 2.1.

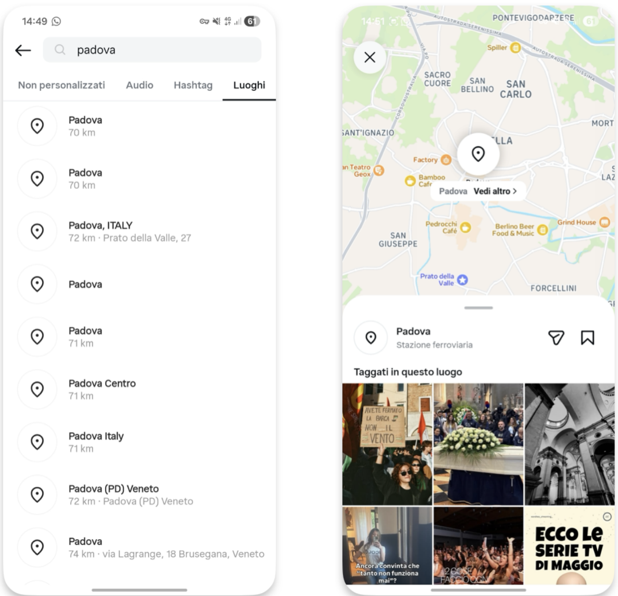

Figure 2.5: Instagram map content 2026

2.4 Google Maps

Google Maps[19] is one of the services provided by Google Inc. Initially released as a web-based platform and following the wide adoption of mobile devices, extended for both iOS and Android systems. The service provides satellite imagery, aerial photography, route planning and 360° interactive panoramic views, commonly known as "Street View"[21]. Google Maps consents users to explore their surroundings by displaying stores, points of interest, and attractions near to their current location or consnet to explorer it scrolling it. The application consents users to explore their surroundings by displaying stores, attractions or others POI POI near their current location, or by enabling them to explore other areas by scrolling the map. These points of interest can be organized into categories such as restaurants, coffee shops, and other predefined types as shown in Figure 2.6. For each place, users may submit reviews by commenting or uploading pictures and videos, thus to create a collaborative information ecosystem. Similarly to Foursquare Swarm 2.1, Google Maps also introduces gamification elements such as badges and user levels to incentivize contributions and improve the quality of place-based reviews.

Google Maps further provides APIs and libraries that allows the development of integrations to its services into thrid-party applications. In fact, the GeoMedia client app 5 through the React Native library adopted, renderized the Google Map as default map framework for Android devices. This service is also well integrated in its ecosystem and Android platforms as well, in fact, exists several standard features, for an instance 'recent places' for places visited by users (within their smartphone) or to have the routing to home or work, enabled and suggested when connected to the car or started in Android Auto [16].

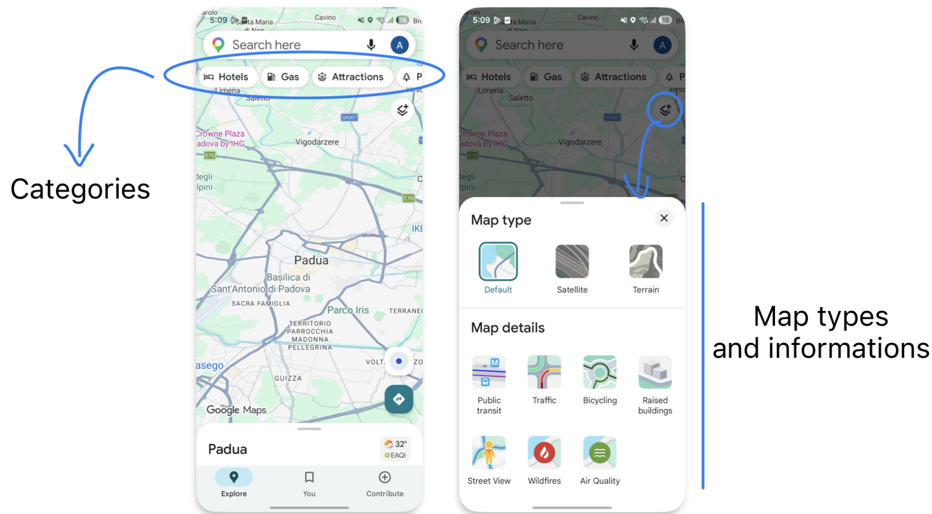

Figure 2.6: Google Maps show case

2.5 Others

There are several other social media platforms that integrate map features and geolocation capabilities. Although they are less popular than the platforms previously discussed, they still cover different scenarios and provide innovative functionalities. For example, Strava, originally developed as an application for tracking running and cycling activities, has evolved into a social networking, by allowing users to share their sports activities, connect with friends, and discover new training partners. While Strava uses geolocation primarily to support sports-related activities, other services adopt location-based interaction at the core of the user experience. More closely related to GeoMedia there are two social media platforms, Jagat [23] and Auglinn [15], both focusing on enabling users to share content and interact with others through a geographical map and current location.

2.5.1 Jagat

Jagat offers several functionalities and highly responsive performance, despite a difficult usability perspective for first-time users, which can appear crowded and confusing, as it displays too many buttons and pieces of information simultaneously, as can be seen in Figure 2.7.
Similar to Snapchat 2.3 or Swarm 2.1, Jagat provide a real-time map enriched with the current location of friends but with more metadata like users' movement speed, or time spent in a specific location or they device battery level. These features raise significant priacy and safety concerns due to the amount of personal information collected and shared.

As social media platform, it also enables users to upload photos and videos, not anchored to any specific location but whenever users want; functionality implemented also by GeoMedia.

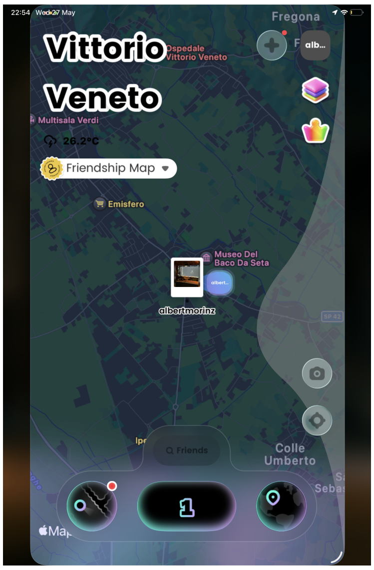

Figure 2.7: Jagatt map interface

2.5.2 Auglinn

Available for both Android and iOS platforms, and also for the web, Auglinn is the closest application to GeoMedia. It provides a platform that allows users to create messages enriched with multimedia content and share them with other users through a geographical map. Moreover, Auglinn also offers augmented reality (AR) functionalities to display location-based messages, which can be created either remotely or on demand through a mobile device. It also allows users to create different public, view-only, or private spaces, which act like containers for content (called “notes” or “boxes”) and set an expiration time for them, as GeoMedia as well. Other users can respectivaly interact by commenting, saving, liking or sharing notes found.

An example of Auglinn usage is shown in Figure 2.8

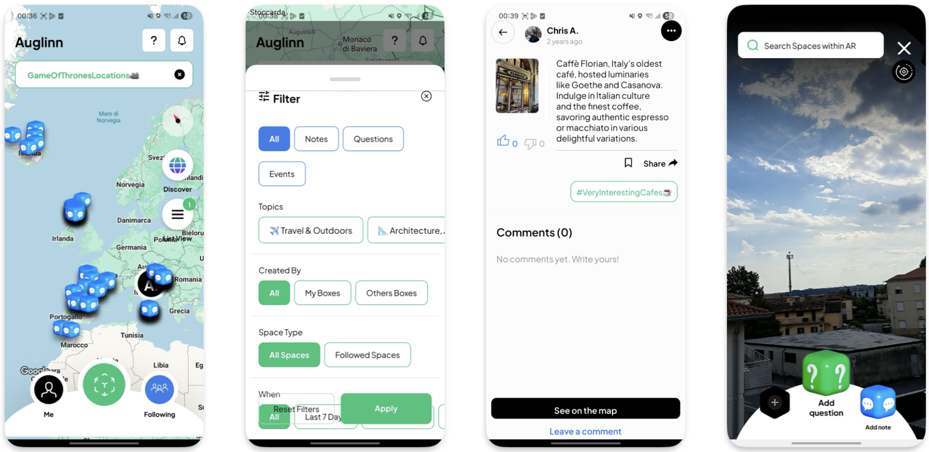

Figure 2.8: Auglinn show case

Chapter 3

GeoMedia

3.1 Concept

GeoMedia introduces a geographically oriented social platform in which users can upload and share content in the form of posts. Each post acts as a container for the information that users want to publish, enabling location-based interactions and content discovery.

The original concept focused exclusively on textual comments (called *GeoComments*), then extended to support any kind of multimedia content. Finally enriched with the concpet of collections, which act as layers for group posts and provide more sophisticated and domain-specific use cases, allowing users to organize and restrict the scope of displayed content.

To realize the idea behind the application, an appropriate software architecture (later explained in Cap. 4) and different supporting technlogies must be adopted. The client layer, which represent the primary point of interaction between users and the platform, is implemented as mobile application. This app allows user to discover the physical world while leveraging GPS capabilities of the device, thust to interact with the application layers and relative posts. Moreover, the mobile app needs to communicate with a backend, which works as coordinator for implementing the system‘s business logic and determining which content should be provided to the users' requests (basing on their current position and other metdata).

3.2 Collections

Collections represent containers for posts that users can create and customize by assigning a title, an icon and a color, allowing them to be easily distinguished from other collections.

Users can also configure the visibility of each collection according to the following dimensions:

- **Users**: specifies which users are authorized to view the collection in the different sections of the application (e.g., the map view and the collections list).
- **Time**: restricts the visibility of the collection to a specific date range. Additionally, recurring visibility intervals can be configured on a monthly basis (repeating on the

specified days of each month) or on a yearly basis (repeating during the specified month and days each year).

Additionally to defining who can view the collection, owners of a collection can specify which users are authorized to create post within it. This distinction between viewers and contributors provides finer-grained access control, enriching users experiences while enabling more effective moderation and content management.

For each collection, the owner can also enable or disable the *remote posting* feature. When enabled, users are allowed to manually select the geographical location associated with a new post, byu dragging a marked through the map. Instead, when disabled, posts can only be created at the user's current GPS location. This last case, makes the platform suitable for a wide variety of scenarios, such as restaurant reviews, traffic reports and real-time local information; avoiding the likelihood of unrelated or misleading content and strengthening the connection between the online content and the physical world.

Moreover, the creator of the collection can reorder the contained posts, enabling a sequential visibility order in which a post becomes visible only after the previous one has been viewed.

Figure 3.1 shows the explained functionalities with the respective order:

- The panoramic view of a single collection (top-left);
- The sequential ordering function (top-right);
- The application of date-time and recurrence limitations (bottom-left);
- The management of creators allowed within the collection (bottom-right).

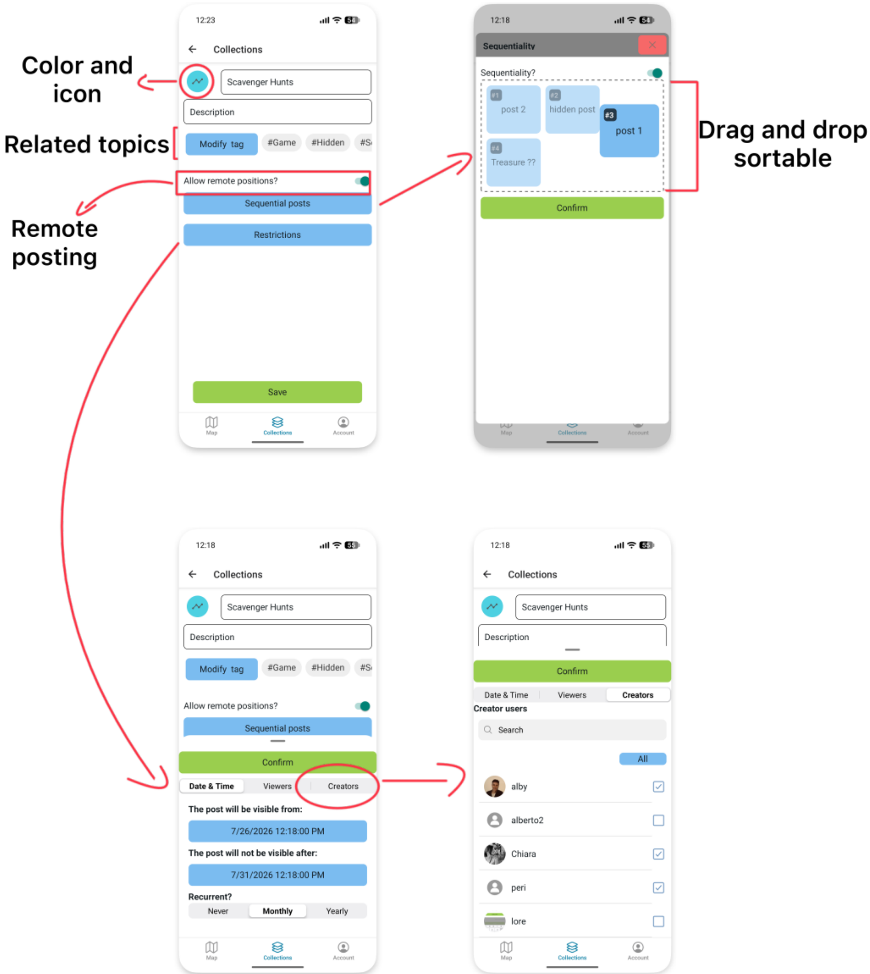

Figure 3.1: Collection panormaic and functionalities.

Finally, as shown in Figure 3.2 each collection can be associated with an unlimited number of hashtags, which serve as thematic labels for content classification. These hashtags facilitate users to discover topics and identify collections of specific interests, toward to create a personalized feed in the "For you" section of collections.

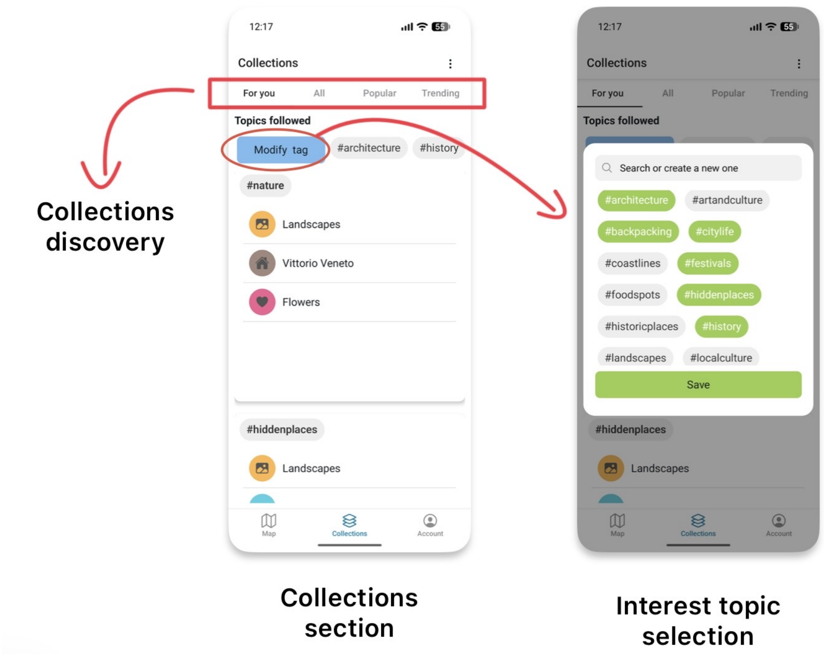

Figure 3.2: Collection topics and suggestions

3.3 Map

The **Map** section represents the core component of GeoMedia, providing an interactive geographical map enriched with location-based posts. Each post is represented by a marker positioned at the geographical coordinates where it was created or, when the associated collection allows *remote posting*, at the location selected by its creator.

To improve visual recognition, each marker adopts the color assigned to its collection, allowing users to associate the context to which the post belongs.
By selecting a marker, the complete content of the corresponding post is displayed and express appreciation by liking the post, or, whenever a post contains an attached media file, users can download it regardless of its file format. Furthermore, users, can view the creator's profile, checking its statistic displayed in graphs or browse other posts of the same creator which are accessible from current location (and others metadata). To preserve users' privacy, posts displayed in viewer mode are represented as lists and rather than on a map, thus to hide the positions of the creator and preserving the gamified experience of discovery through the map. However, this feature provide a peek of some content of the user in way to provide users potential related posts.

As shown in Figure 3.3, the interface (described in Chapter **??**) has been designed to provide a simple and intuitive user experience. In particular, it highlights the most relevant collections available around the user's current location by prioritizing those containing the largest number of posts near the current location, this feature is called "local collections". Moreover, GeoMedia offer different modalities toward the discovery of collections, such as:

- **For you**: That provide a personalized feed for user, to discover collections based on their interests topic selected within the same page.

- **All**: Which provide all the collections in form as a list in random order.
- **Popular**: Represents a ranking of the most frequently used collections, computed based on the total number of posts associated with each collection.
- **Trending**: Represents a ranking of the most frequently used collections during the last four hours.

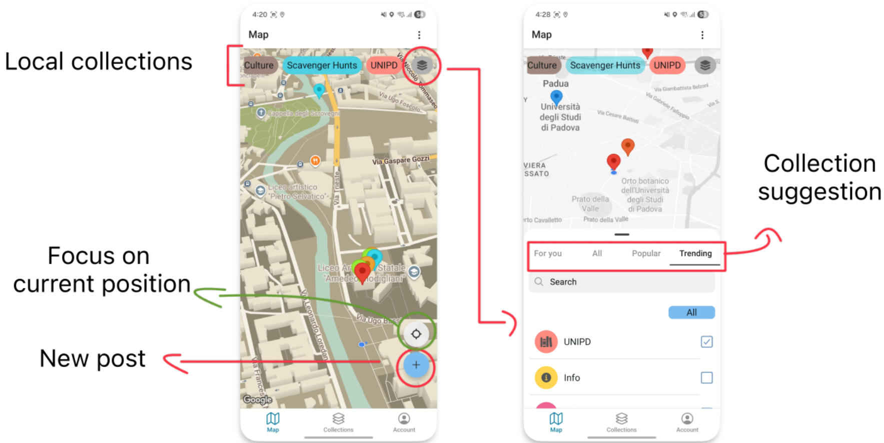

Figure 3.3: Map tab overview and colletion discovery

GeoMedia also supports multiple styles among five different map themes, through the application settings, allowing the interface to better adapt to personal preferences and different usage scenarios. The current position of the user is continuously followed through the GPS capability thus to make appear and hide different posts as user moves through the physical world; then the map automatically shift its center to the current location (unless the user manually navigates the map).

3.3.1 Post creation

Through the map component, users can also access to the post creation section, where the following required information are presented:

- **Collection**: the collection to which the post will belong.
- **Title**: the title of the post.
- **Visibility Range (km)**: the geographical distance within the post is visible to other users (between *0.2 - 30.0 KM*).
- **Description (optional)**: a textual description with no practical limitation on its length.
- **Media (optional)**: allows users to attach one or more files from the mobile device, including images, videos, audio recordings or any other supported file type. Alternatively, the integrated camera can be used to capture a photograph and share it within the post.

- **Visibility Constraints**: optional restrictions that limit the accessibility of the post according to temporal conditions (a date-time interval with optional recurrence) or user-based permissions (specifying which users are allowed to access the post).

The geographical position of a post is automatically retrieved from the user's current GPS location. Each post is therefore characterized by its latitude, longitude, and altitude. Although altitude is not currently exploited by the platform, it is stored to support future extensions of the system.

3.3.2 Haversine formula

To determine which posts are visible to a user, GeoMedia evaluates the geographical distance between the user's current position and the location associated with each stored post. Only posts whose visibility radius includes the user's current position are returned by the server and displayed on the mobile devices.

The server computes these distances using an implementation of the Haversine formula[45]. Specifically, the implementation, reported in Listing 3.1, calculates the great-circle distance between two points on the Earth's surface from their latitude and longitude coordinates. The resulting distance is then compared with the visibility radius associated with each post to determine whether the post is within the user's area of visibility.

The Haversine formula models the Earth as a sphere and provides an efficient approximation of the distance between two geographic coordinates. Although ellipsoidal models such as Vincenty's formulae[55] provide higher accuracy, overral the precision offered by the Haversine formula is sufficient for this application, where the visibility of a post depends on whether a user is located within a predefined proximity range. Furthermore, its low computational complexity makes it particularly suitable for backend applications requiring frequent geographical distance calculations.

For short-range distance calculations, the error introduced by the spherical approximation is negligible compared with the uncertainty of standard GPS positioning. At distances of approximately 30 km, the deviation from an ellipsoidal model is generally within a few meters, while for distances below a few kilometers it is typically reduced to the meter or sub-meter range.

```
# Haversine formula implementation
# Compute the geodesic distance between two geographic coordinates
# using Vincenty's inverse formula.

# Args:
#     areaKM (float): The range within the post would be visible
#     post_lat (float): Latitude of the post.
#     post_lon (float): Longitude of the post.
#     curr_lat (float): Current latitude coordinate of the user.
#     curr_lon (float): Current longitude coordinate of the user.

# Returns:
#     bool: True if the user position is included within the area of visibility of
    the post.
```

```python

def check_post_area(areaKM, post_lat, post_lon, curr_lat, curr_lon):
    from math import cos, asin, sqrt, pi

    post_lat = float(post_lat)
    post_lon = float(post_lon)
    curr_lat = float(curr_lat)
    curr_lon = float(curr_lon)


    dLat = (post_lat - curr_lat) * pi / 180
    dLon = (post_lon - curr_lon) * pi / 180

    a = (0.5
        - cos(dLat) / 2
        + cos(curr_lat * pi / 180)
        * cos(post_lat * pi / 180)
        * (1 - cos(dLon)) / 2
    )

    d = round(6371000 * 2 * asin(sqrt(a)))

    return d <= areaKM * 1000
```

Listing 3.1: Python implementation of the Haversine-based visibility check.

3.4 Exclusivity

The exclusivity feature assume an important role in GeoMedia, as it allows the application to be adapted to different use cases and real-world scenarios by providing fine-grained control over content visibility.

As previously described, collections act as containers for posts. Consequently, posts inherit the visibility constraints defined at the collection level and may additionally introduce narrower restrictions. In other words, if a user cannot access a collection, they cannot access any of the posts contained within it. However, access to a collection does not automatically imply access to all its posts, as individual posts may apply additional limitations as:

- **Geographical range**: each post defines a visibility radius, which determines the maximum distance from its location at which it can be accessed. The supported range goes from 0.2 km up to 30 km.
- **Collection-based limitations**: posts inherit the restrictions defined by their parent collection, including:
  - **User limitation**: defines which users are allowed to access the collection.
  - **Time limitation**: defines the period during which the collection is visible, consequently the possibility to access to the post.

- **Post-specific time limitation**: applies temporal restrictions directly to a single post, narrowing the visibility inherited from the collection.
- **Post-specific user limitation**: restricts access to a single post for a specific group of users, narrowing the permissions inherited from the collection.
- **Sequentiality limitation**: collections that enable sequential access expose only the next available post (or the first post when starting the sequence) according to the order defined by the creator. This mechanism allows creators to guide users through a predefined path or experience.

3.4.1 Use cases

By combining geographical positioning, collections, access restrictions and sequentiality, GeoMedia supports different scenarios and applications. Some examples are described below:

- **Local tours**: Through collections, which act as containers for multiple related posts, users can select specific layers created by a guide or organization and access only the content belonging to that experience. Furthermore, the sequentiality feature can enrich guided tours by introducing a predefined order, allowing the next attraction or point of interest to become available only after the previous one has been visited.
  Posts can be created remotely by different authors (such as professional guides) and promoted across related collections through the use of hashtags. As a further development, this functionality enable a monetization mechanisms for the platform, allowing creators to provide exclusive paid experiences directly through the platform.
- **Recurrent posts**: GeoMedia allows users to define post visibility not only through geographical constraints but also through temporal conditions. A post can be associated with a specific date and time interval and can optionally be configured with a recurrence pattern, such as monthly or yearly availability. This enables scenarios where the same content becomes available periodically without requiring manual recreation, relevant for scenario such as anniversaries, birthdays or recurrent celebrations.
- **Sequential posts**: Collections with sequentiality enabled allow creators to define a specific order in which posts should be discovered. Only the first available post is initially displayed, while subsequent posts become accessible only after seen the previous one and satisfyingvisibility conditions defined by their geographical position, time constraints, and user permissions.
  This mechanism introduces gamification possibilities within the social platform, enabling experiences such as digital scavenger hunts, interactive tours, treasure hunts or location-based challenges, where users progressively unlock new content by exploring the real world.
- **Augmented Reality**: GeoMedia does not currently implement Augmented Reality (AR) functionality; however, it includes an in-app camera component that can be leveraged for future AR-based enhancements. This feature could enable the integration of digital information above the real-word environment, visualizing for example historical information or media content. With this development, the mobile application would

improve the user experience by providing a more immersive and interactive way to explore and interact with geographically referenced content.

An interesting scenario can be adopted by the City of Padua, thus to illustrate through the app one of its well-known sayings: "città dei tre senza" (literally, "the city of the three withouth"). This expression refers to three notable things that Padua has-but or hasn't in a metaphorical way:

- **The lawn without grass**: referring to *Prato della Valle*, which despite its name ("meadow of the valley"), feature ornamental grass and is not a lawn in the traditional sense;
- **The saint without a name**: referring to *Sant'Antonio da Padova* who was actually born in Portugal, and only spent the last years of his life in Padua.
- **The café withouth doors**: referring to one of the most historic Italian café, *Caffé Pedrocchi*, which during wartime remained with doors opened all day and night to provide shelter.

In this case, digital markers at these location as shown in Figure 3.4, enable the storytelling of each unique characteristic, by uploading historycal photograph or facts.

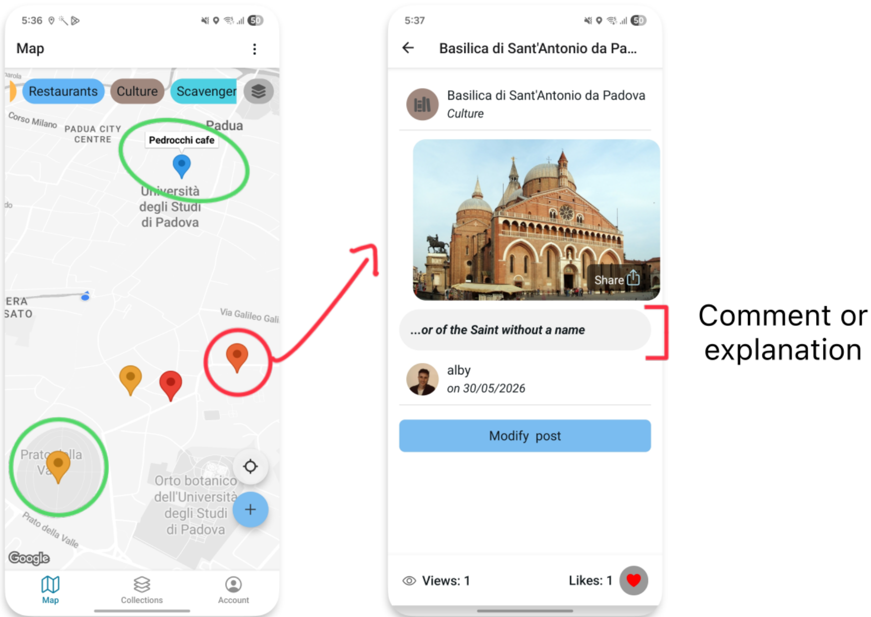

Figure 3.4: City of Padua use case for its sayng of "Tre senza".

Moreover, can

3.5 User

Even if not profile-centric, GeoMedia, allows users the ability to personalize their profiles by specifying basic personal information, such as their first and last name but also to upload a profile picture. These features allow users to establish a recognizable identity within the platform and enhance the overall user experience.

In addition, the GeoMedia mobile application provides users with infographic-based statistics that summarize their activity on the platform, such as the number of published posts distribuited across months or which in which collections they are active as pie-chart. As shown in Figure 3.5, these visual summaries offer users an immediate overview of their contributions and engagement within the application.

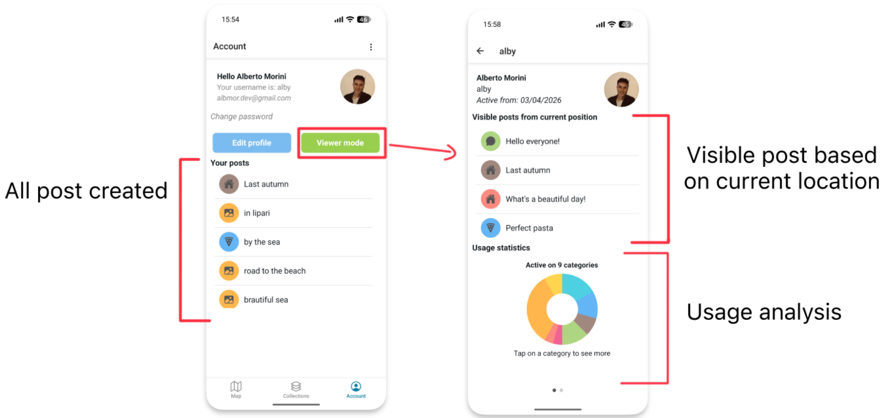

Figure 3.5: User profile editor and viewer mode.

3.6 Functional comparison

While GeoMedia incorporates functionalities commonly found in existing social media platforms presented in Chapter 2, it extends their capabilities through several distinctive features. Table 3.1 provides a comparative overview of the supported functionalities across the analyzed platforms.

Feature	GeoMedia	Jagat	Aulinn	Foursquare Swarm	Snapchat	Instagram	Google Maps
Remote location posting	✓	✓	✓	✓	–	✓	✓
Location-based restrictions	✓	–	–	–	–	–	–
Time restrictions	✓	–	–	–	✓	✓	–
Sequential posts	✓	–	–	–	–	–	–
Multi-author collections	Selective contributors	–	–	–	–	Everyone	Everyone
User-based restrictions	Selective viewers	–	–	–	Selective viewers	Selective viewers	–
Topic-based content	✓	–	–	✓	–	–	✓
User statistics	✓	–	–	✓	–	–	✓

Table 3.1: Functional comparison of GeoMedia and related social media platforms.

Chapter 4

Architecture and Software design

The software architecture to concretize the concept behind GeoMedia, is represented by a three tier solution, which divide the whole platform into different actors accordign they operativity and functions.

In specific, a three tier model is costituited by:

Generally, a three-tier architecture consists of the following layers:

- **Client layer (Presentation layer):** which provides the user interface and manages user interactions. It is responsible for presenting data to the user and forwarding user requests to the application layer while remaining independent of the underlying business logic and data storage. *In GeoMedia, this is represented by the mobile app*
- **Application layer (Business Logic layer):** that implements the core functionalities of the platform and applies the business logic to process user requests. This layer acts as an intermediary between the presentation and data layers, encapsulating the application's logic and coordinating the exchange of data to ensure consistency and proper execution of system operations. *In GeoMedia, this layer is represented by the server-side component responsible for processing client requests, executing the application logic and coordinating communication between the client and the data layer through a RESTful interface.*
- **Data layer (Persistence layer):** that is responsible for managing data storage and performing data operations such as the creation, retrieval, update and deletion (CRUD) of persistent data. It provides an interface to databases or other storage systems while abstracting and encapsulating the underlying data access details from the upper layers. *In GeoMedia, this layer is represented by the Database Management System (DBMS) that provides a full solution for data handling.*

This solution provide a classic and simple architecture, by exploiting technology functionalities such Network communication with TCP/IP stack, these actors are able to enstablish a communication among each other and then to exchange information and activate business process.

The UML schema in the Figure 4.1 can represent graphically the architecture enstablished.

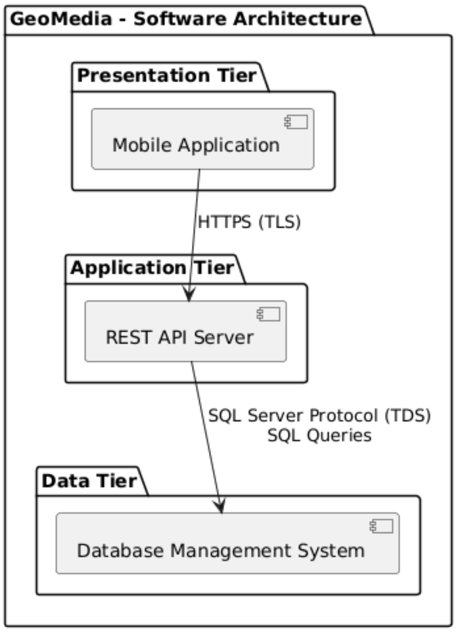

Figure 4.1: Three-tier Software Architecture UML.

Into the detail of GeoMedia, the communication is implemented through the HTTP protocol, which is one of the most popular and widely used protocols in the last decades. HTTP is a request-response protocol based on the client-server model that create a virtual link between the application layer (as server) that returns a response to the client (presentation layer) with the data requested. The data, referred to as packets, can be manipulated through intermediary network nodes (proxy servers, NAT, web caches, etc.) while respecting the TCP/IP structure. In order to ensure the confidentiality and security of the data, HTTP allows the adoption of SSL certificates (HTTPS), making the communication encrypted with the public key of the server (and decryptable only with the sercret key avaiable only to the application layer).

Instead, to enstablish the connection between the others layer (application and data) the protocol depends on the data layer, in the case of GeoMedia the DBMS is represented by Microsoft SQL Server (MSSQL) that operate through the TCP/IP stack with TDS (Tabular Data Stream) protocol.

4.1 Microservices comparison

Another valid alternative to the architecture design adopted is the microservices structure, that see the system decomposed into a set of small and independent services, each responsible for a specific business capability.

Each microservice encapsulates its own business logic and expose its functionalities through interfaces, usually providing RESTful APIs. In fact, this design concenrs mainly the backend layers such as application and data more than the client which still represent the single poin of interaction with the users.

This design compared to the a monolithic solution provides high degree of modularity and flexibility and others several key advantages, such as:

- **Scalability:** individual services can be scaled independently according to their workload, reducing infrastructure costs and improving resource utilization.
- **Maintainability:** the decomposition into smaller services simplifies both code maintenanc than testing and debugging, as each service has a limited and well-defined responsibility.
- **Technology independence:** each service can be implemented using the programming language, framework or database that best fits its operativity, while respecting the compatibility thorugh communication interfaces
- **Fault isolation:** failures occurring in one service are less likely to compromise the entire application, increasing the reliability of the system.
- **Continuous deployment:** updates can be released independently for each service, enabling faster development cycles and reducing deployment risks.

Despite these advantages, the adoption of a microservices architecture represent an additional complexity which can be avoided for simple projects or applications with low usage. In fact, the challenge related to network communication, together with issues such as data confidentiality and authentication, distribuited data transactions or service orchestration tend to outweight the architectual benefits.

The three-tier architecture designed for GeoMedia represents a monolithic solution that is sufficient for the current size and usage of the application, successfully supporting its functional requirements. Nevertheless, the clear separation between the presentation, application and data layers ensures that the software remains extensible and can progressively evolve towards a microservices architecture without requiring a complete redesign.

On the other hand, the adoption of a microservices architecture could be an interesting and justified scenario to support local and distributed databases across different countries. Since GeoMedia focuses on local data, synchronization would only be required for user accounts and collections, while posts could be stored on country-specific servers according to their geographical coordinates metadata.

A concept of the microservice adoption for GeoMedia is shown in the UML in Figure 4.2

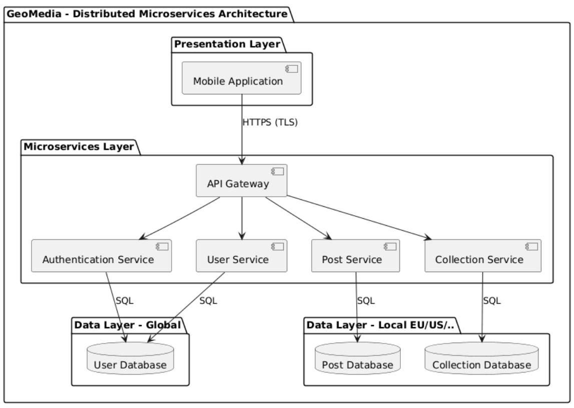

Figure 4.2: Microservices Software Architecture UML.

Chapter 5

Mobile app

In order to provide its core functionalities, GeoMedia relies on mobile devices, such as smartphones and tablets, where GPS capabilities and internet connectivity can be leveraged to enable the discovery of the surrounding environment. Through the mobile application, users can visualize nearby posts based on their location and contribute new content by capturing images using the integrated camera. Therefore, the mobile application represents the most important component within the entire framework. Furthermore, its development introduces additional design challenges related to User Experience (UX), usability, performance limitations of mobile devices and the protection of user privacy.

5.1 React Native

React Native[47], commonly abbreviated as RN, is an open-source software development framework created by Meta Platforms (formerly Facebook Inc.) for building cross-platform mobile and lately even desktop applications. Unlike traditional development approaches, which require separate codebases for different operating systems, React Native enables developers to maintain a single codebase that is transformed and adapted during the deployment phase toward the destiantion platform. This approach allows the reuse of all or at least the most of the application logic across different platforms, such as Android and iOS. Consequently, it reduces development and maintenance efforts while preserving the performance and user experience expected from native applications.

Under the hook, React Native relies on a component-based architecture, where user interfaces are defined through reusable React components. React itself is a framework originally designed for web application development, introducing an efficient rendering mechanism that updates only the components affected by changes in application state, avoiding unnecessary re-rendering operations. React Native extends this concept by enabling communication between Javascript/TypeScript based components and native platform APIs, allowing applications to access specific peripherals like GPS, cameras, haptic feedback actuators or several other embedded sensors and functionalities. More specifically, React Native consents to write application logic using web-oriented programming languages, such as JavaScript or TypeScript, while adopting a styling approach similar to CSS for interface design. UI elements are represented through reusable components, which can be considered analogous to HTML elements in traditional web development. Finally, during the deployment process, these components are translated into their corresponding native platform components, this enable the app to achieve a native-like user experience while maintaining a shared development environment.

For GeoMedia, React Native represents a suitable technological option, allowing the platform to reach both Android than iOS markets. Overral, React Native is well supported by various components and libaries that can fully satisfy the requirements of the service.

5.1.1 Ionic Framework

A first version of GeoMedia was been developed with IonicFramework[38], which is a valid framework for webdevelopment with the capability to encapsulating the web application into a native one that will render and interpretate the content as browsers do.

Moreover, as advantages, Ionic allows developer to develop with different web framework such as: ReactJS, AngularJS, VueJS and integrate the Ionic components.

A first version of GeoMedia was developed using Ionic Framework[38], which is a valid framework for web development with the capability of encapsulating web applications into mobile applications. This approach could seem to be similar to ReactNative but it's core is totally different, since IonicFramework rerender and interpretate the components on running time, as web browsers process web pages, while ReactNative compile and traduce into native components providing higher performance and more optimal resource utilization.

Moreover, one of the main advantages of Ionic is the possibility for developers to work with different web frameworks such as ReactJS, AngularJS or VueJS, while integrating Ionic's components.

A visual comparison between the Ionic previous version of GeoMedia and the React Native final version is shown in Figure 5.1.

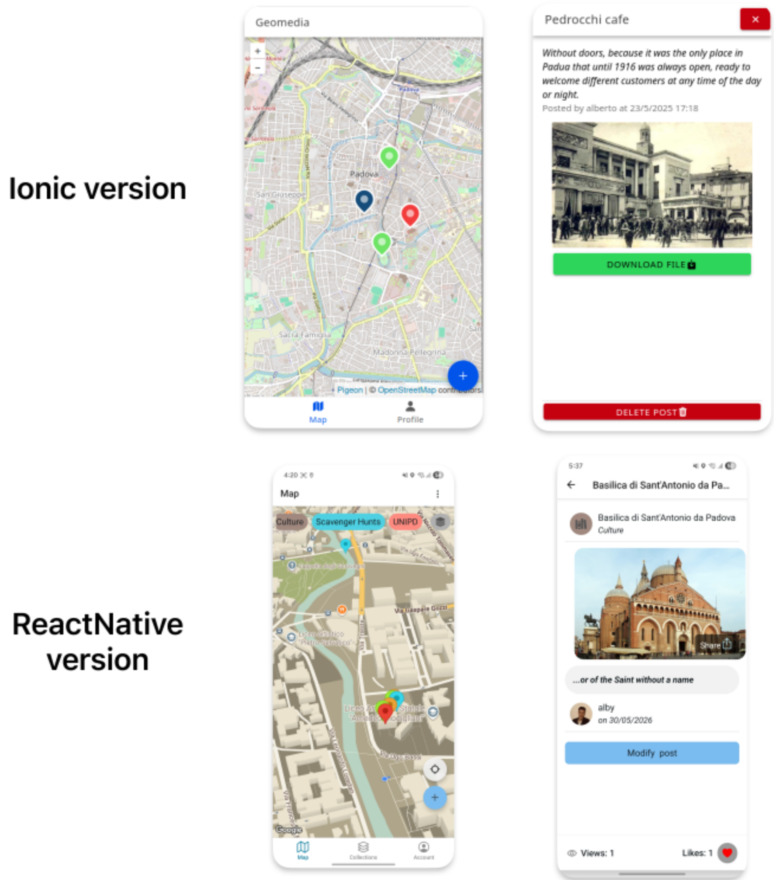

Figure 5.1: GeoMedia UI comparison between the Ionic Framework version (above) and the React Native version (below).

5.1.2 Libraries

One of the main strengths of React Native is its extensive ecosystem of free and open-source libraries and components. These resources are widely documented and maintained by the community, with the official React Native Directory [46] serving as the primary reference for discovering and evaluating compatible packages.

Overral GeoMedia for the Android version developed use several dependencies packages such as:

- **@gorhom/bottom-sheet**: provides customizable bottom sheet components that can be expanded or collapsed to display additional content without leaving the current screen, used mainly in the map component to render collections.
- **@react-native-community/geolocation**: provides access to the device's geolocation services, enabling the application to retrieve the user's current position.
- **@react-native-community/slider**: offers a slider component that allows users to select value within a specified, applied for chose the distance range in the post visibility.

- **@shopify/flash-list**: offers a high-performance list component optimized for rendering large collections of data with improved memory usage and scrolling performance.
- **expo**: provides the development framework and runtime environment for building, testing and deploying React Native applications, together with a wide range of native APIs.
- **expo-router**: implements a file-based routing system that simplifies navigation between the different screens of the application.
- **expo-secure-store**: securely stores sensitive information, such as user credentials, by leveraging the secure storage facilities provided by the underlying operating system.
- **react-native-gesture-handler**: enables advanced gesture recognition, including swipes, taps and drag interactions, providing a more responsive and native user experience.
- **react-native-gifted-charts**: provides customizable chart components for visualizing statistical information about the usage on the user profile.
- **react-native-maps**: integrates interactive maps into the application, allowing the visualization of geographic information and user locations.
- **react-native-reanimated-carousel**: implements an animated carousel component for displaying file attached into a post through horizontal scrolling.
- **react-native-share**: through the native sharing facilities of the operating system, allows to share or store the media attached in a post such as images or documents-
- **react-native-sortables**: provides sortable list components that allow users to reorder items through drag-and-drop interactions, used to arrange the post whenever th esequentals features is active in a collection.
- **react-native-vision-camera**: provides high-performance access to the device's camera, enabling image capture thus to attach piuctures in the post withouth levaing GeoMedia.

A particular emphasis should be placed on libraries that provide access to device APIs and hardware features, such as sensors and other physical components.

Map & GPS

The map is the core component of GeoMedia and is implemented using the 'react-native-maps' library, which during the compilation process is translated into the corresponding native implementation for each platform, namely Google Maps on Android and Apple Maps on iOS.

On the Android platform, the Google Maps component provides a wide range of features. These include graphical customization of the map through styling properties, as well as the display of commercial, historical or other points of interest, such as coffee shops, restaurants or retail stores. In GeoMedia, these elements remain visible and serve as useful landmarks, helping users orient themselves within the displayed environment and among the posts published through the app itself. In fact, the map supports the rendering of additional geographical elements with specific coordinates, known as markers. Each marker acts as a visual placeholder that identifies the physical location associated with a post.

In addition, the library exposes several callback functions and hooks that allow developers to implement custom logic. One such feature is continuous background user location tracking, which GeoMedia leverages to dynamically load location-aware posts as the user moves through the physical environment. This approach ensures that the displayed content remains relevant to the user's current location. To optimize network usage and avoid unnecessary server requests caused by minor location variations, GeoMedia applies a 50-meter movement threshold, triggering a new request and post update only when the user's displacement exceeds this distance.

In order to access Google Maps services on Android, a Google Maps API key is required, which serves as an authentication token that links the application to a Google Cloud Project (a Google suite that offers access to other products such as shared services, computing resources, and others), enabling access to the Google Maps SDK and its associated services. The API key follows the pricing policy defined by Google. Specifically, it applies a 'pay-as-you-go' pricing model[11], in which the first 10,000 map loads (renderings) per month are free, while each additional 1,000 map loads costs 7 $.

An implementation of map renderig is shown in Listing 5.1

```
<MapView
        ref={mapRef}
        style={styles.map}
        showsUserLocation={true}
        toolbarEnabled={true}
        followsUserLocation={true}
        initialRegion={{
            latitude: UserPosition.latitude,
            longitude: UserPosition.longitude,
            latitudeDelta: 0.1,
            longitudeDelta: 0.1,
        }}
        customMapStyle={
            mapPreferenceStyle == "system" ? //here thus to render automatically
    when toggled by user via status-bar
                colorScheme == "dark" ? MapDarkMode : MapLightMode
                :
                mapPreferenceStyle
        }
        onUserLocationChange={(event) => {
            const { latitude, longitude } = event.nativeEvent.coordinate;

            if (positionChanged(latitude, longitude)) {
                setUserPosition({ latitude: latitude, longitude: longitude });
                get_posts_map({ latitude: latitude, longitude: longitude })
    //retrieve new posts
```

```
            }
        }}
        zoomEnabled={true}
        camera={{
            center: {
                latitude: UserPosition.latitude,
                longitude: UserPosition.longitude,
            },
            pitch: 30, // like the angle
            heading: 0, // direction the camera faces (0 = north)
            zoom: 13.7, // optional, overrides altitude
        }}
        pitchEnabled={true}
        showsCompass={false}
        showsBuildings={true}
        showsMyLocationButton={false} //make it custom thus to give space to collections
    >
        {//rendering the markers
            postMarkers?.map((p, index) => (
                <Marker
                    key={p?.ID}
                    coordinate={{
                        latitude: p?.LATITUDE,
                        longitude: p?.LONGITUDE,
                    }}
                    pinColor={p?.COLOR}
                    title={p?.TITLE}
                    onPress={() => { ///redirect to post viewer
                        router.push({
                            pathname: 'viewer/PostViewer',
                            params: {
                                postid: p?.ID,
                            }
                        });
                    }}
                />
            ))
        }
    </MapView>
```

Listing 5.1: Implementation of MapView component for rendering posts.

Along with the map rendering, the integration of the GPS module provides the current coordinates of the user:

- **Longitude**: the angular distance east or west of the Prime Meridian, used to identify the horizontal position on the Earth's surface.
- **Latitude**: the angular distance north or south of the Equator, used to identify the

vertical position on the Earth's surface.

- **Altitude**: the vertical distance of the user above mean sea level, estimated by the GPS sensor.

Even though altitude is not currently used by GeoMedia, it is stored for future developments.

This set of information is repeatedly used throughout GeoMedia, both for displaying nearby posts and during the post creation process, where the user's current position is automatically assigned as the post location. Specifically, when the current position is used, the altitude is also stored. Conversely, when the location is selected remotely, the altitude is not available. This is because the altitude can be estimated by the GPS sensor, whereas the altitude of a manually selected location cannot be reliably determined, as different points may correspond to buildings with multiple floors, skyscrapers, or mountainous areas.
In order to access device resources and user data, Android (as well as the iOS platform) adopts a permission-based security model. This requires users to explicitly grant permissions before applications can access protected system features, hardware components, and sensitive information.

An implementation of position retrieving is shown in Listing 5.3

```

async function getLocation() {
       const hasPermission = await requestLocationPermission();

       if (!hasPermission) {
           Alert.alert(langselected?.permission.location.denied);
           return;
       }
       if (Platform.OS === 'ios') {
           Geolocation.requestAuthorization();
       }

       Geolocation.getCurrentPosition(
           position => {
               const { latitude, longitude } = position.coords;
               if (positionChanged(latitude, longitude, UserPosition)) {
                   setUserPosition(prev => ({ ...prev, latitude: latitude, longitude: longitude }));
                   get_posts_map({ latitude: latitude, longitude: longitude })
               }

               // set map with center on user location
               mapRef.current?.animateToRegion({
                   latitude,
                   longitude,
                   latitudeDelta: 0.03, //zoom
                   longitudeDelta: 0.03, //zoom
               }, 1000);
               return { latitude: latitude, longitude: longitude }
           },
```

```
        ...

}
```

Listing 5.2: Location retriving snippet.

File sharing & camera access

Another external implementation involving device sensors is camera access. This functionality, which is always granted through the permission model, allows GeoMedia to access the smartphone camera capabilities in order to capture images and upload them during the post creation process.

To accomplish this objective, GeoMedia integrates the third-party library 'react-native-vision-camera'. This library provides a set of hook functions that encapsulate the underlying camera access logic and simplify the management and storage of captured images, as shown in Listing **??**.

```

export default function OpenCamera(
    { onClose, storePhoto }: { onClose: () => void; storePhoto: (b64: string) =>
    void }
) {
    ...
    return (
        <View style={StyleSheet.absoluteFill}>

            <Camera
                ref={camera}
                style={StyleSheet.absoluteFill}
                device={device}
                isActive={true}
                photo={true}          // enable photo capture
                video={false}         // enable if you want video later
                enableZoomGesture={true}
                torch={flash === 'on' ? 'on' : 'off'} // continuous torch (flash
    preview)
            />

            {/* FLASH/REVERSE BUTTONS */}
            ...

            {/* CAPTURE BUTTON */}
            <View style={styles.controlsBottom}>
                <TouchableOpacity style={styles.captureButton} onPress={async ()
    => {
                    let b64 = await takePhoto();
                    storePhoto(b64)
```

```
                }}>
                    <View style={styles.captureInner} />
                </TouchableOpacity>
            </View>
        </View >
    );
}
```

Listing 5.3: Location retriving snippet.

Alternatively, users can upload different types of files without restrictions on their respective extensions. GeoMedia supports various file formats, including audio files and PDF documents. To enable this functionality, the application allows users to select files directly from the device file system through libraries that interact with the operating system using well-defined system APIs. For this functionality, no specific visual components are required; instead, the file selection mechanism is triggered programmatically through a method call associated with an arbitrary button, which invokes the native file selection interface provided by the platform, as shown in Listing 5.4.

```

    async function file_pick() {
        try {
            const result = await DocumentPicker.getDocumentAsync({
                type: "*/*",
                multiple: true,
                copyToCacheDirectory: true,
            });

            if (result.canceled) return;

            let files = []
            for (const asset of result.assets) {
                const { name, uri } = asset; //filename and uri path
                const mimetype = asset?.mimeType;

                //BASE64 computing
                const file = new FileSystem.File(uri); //load the file
                const base64 = await file.base64();

                files.push({
                    FILENAME: name,
                    UPDATED: true,
                    BASE64: base64,
                    MIME_TYPE: mimetype
                })
            }

            setFilesAttached(prev => [...prev, ...files])

```

```
        } catch (error) {
            console.error("Error picking or reading file:", error);
            Alert.alert("Error reading files", error)
        }
    }
```

Listing 5.4: File picking snippet.

During the visualization of a post, users can interact with the content by performing actions such as adding likes and browsing uploaded files through a carousel view. Furthermore, the sharing functionality allows users to distribute the post content through external applications by exploiting the Android Intent mechanism, which enables communication between applications by describing the type of operation and data to be handled. The Android system then identifies the applications compatible with the requested content through their registered Intent filters, as shown in Figure 5.2.

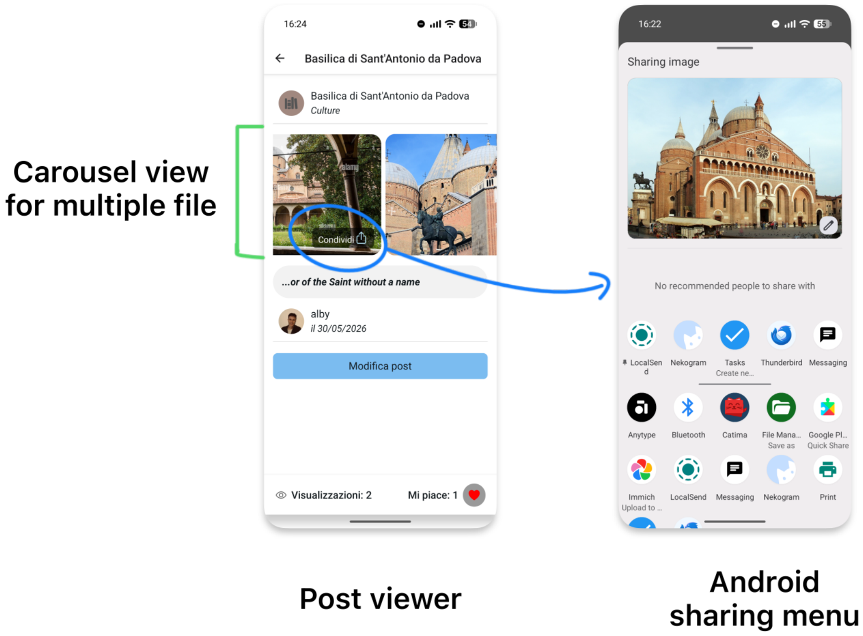

Figure 5.2: Post visualization and Android sharing menu

Thus, to upload captured images or selected files to the server, the content is encoded from its binary representation into Base64 format. This encoding provides a standardized method for representing binary data as a textual format, allowing files to be safely transmitted through protocols and systems that handle text-based data. On the application side, the server will decode and store files into the File System of the server.

5.2 App flowchart

To provide a comprehensive overview of the application structure and user interactions, a workflow diagram of GeoMedia is presented in the Figure.

This diagram describes the main operations performed by users and the corresponding processes executed by the application, highlighting the interaction between the user interface, internal components, and external services. The represtation of the application flowchart clarify the sequence of actions and the dependencies between differnet functionalities expecially during app design process.

5.3 UI & UX

make reference to Lorenzo and Gaggi's paper: https://ieeexplore.ieee.org/document/10454681

5.3.1 Sequential feature

Gestures?

5.3.2 SUS Questionaire results

closing soon

5.4 Android application

Android is an operating system developed by Google, originally designed for mobile devices such as smartphones and tablets, and later extended to a wider range of platforms, including televisions, vehicles, smartwatches, and other embedded systems. It is based on a modified version of the Linux kernel and was first released in 2008. Since then, it has become the most widely adopted operating system for smartphones worldwide[8].

Android provides a high degree of customization, allowing manufacturers such as Samsung, Motorola o others to modify the user interface and include proprietary applications as part of their devices. Despite these customizations, Android remains an open platform that allows users to install applications from several different stores or, more brutally through APK file, that contain the package and executable code to install the app.

Android application development is freely available, with the main exception for deployment phases to in the official Google stores, where developers need to make one-time payment for a developer license that enable them to upload as many apps they want. Furthermore, Android development is platform-independent, representing a significant advantage over competing platforms such as Apple, where application compilation and deployment require dedicated software tools available only on macOS (Apple Inc operating systems for PC).

5.4.1 Development

The development of GeoMedia, since it's a React Native project see different tools cooperating through a well defined sequences in order to develop and build the final application (APK).

The official React Native documentation recommends Expo as the default starting point for most applications, as it provides a production-ready framework with integrated tooling, while still allowing the transition to native workflows when advanced platform-specific functionalities are required.

To develop a React Native application using the Expo framework, the development environment must satisfy a number of software prerequisites:

- **Node.js**: provides the JavaScript runtime environment required to execute development tools and includes the *npm* package manager.
- **npm**: the default package manager for Node.js, used to install and manage project dependencies.
- **Expo**: the framework used to simplify the development, testing, and deployment of React Native applications. It is installed and managed through *npm* and is typically initialized using the `create-expo-app` utility.

Once the prerequisites have been installed, a new Expo project can be created by executing the command `npx create-expo-app@latest`. The application can then be launched on a target platform using the command `npx expo run:$platform$`, where the platform is `android` or `ios`.
For Android, the application can be executed either on a virtual device (through emulator) or physical if debugging mode is enabled.

The generated project directory contains the application source code, static assets, and several configuration files, including:

- **app.json**: stores the Expo application configuration, including the application name, version, icons, splash screen and platform-specific settings.
- **package.json**: defines the project's metadata, scripts and JavaScript dependencies required by the application.
- **tsconfig.json**: specifies the TypeScript compiler configuration, including compiler options and preferences.

5.4.2 Deployment

The deployment phases can follow different roads, from sharing the apk to uploading across differents store those will apply some rules and policy, in specific for GeoMedia has been taken in consideration these platforms

- F-Droid: ....
- Google Play Store: where the policy to upload an app are...
- Obtanium

The deployment phase can be carried out through different distribution channels, starting from the direct sharing of the application package (APK) to publication on dedicated application stores. Each application stores adopts its own policies, technical requirements and review procedures that must be satisfied before an application can be made available to end users.

For the GeoMedia application, the following distribution platforms were considered:

- **F-Droid**: an open-source Android application repository that distributes only Free and Open Source Software (FOSS). Applications published on F-Droid must comply with its transparency and open-source requirements, making it a suitable distribution channel for projects released under open-source licenses.
- **Google Play Store**: the official Android application marketplace maintained by Google. Publishing an application requires the creation of a Google Play Developer account, compliance with Google's Developer Program Policies and finally the review of the application along automated processess.
  Furthermore, for newly registered personal developer accounts, Google requires a closed testing phase involving at least **12 testers for a minimum of 14 consecutive days** before production access can be requested.
- **Obtainium**: an alternative application manager that allows users to install and update applications directly from their official release sources, such as GitHub repositories, without relying on a centralized application store. This approach enables rapid distribution while preserving direct control over application releases.

Initially, GeoMedia was distributed through a direct download link to the application's GitHub Releases page, allowing users to download and install the APK independently. This distribution approach was chosen to simplify the deployment process and facilitate early testing without requiring users to participate in Google Play's closed testing program or to install alternative application stores, such as F-Droid. Furthermore, distributing the application directly through GitHub enabled rapid release of new versions while maintaining full control over the deployment process.

5.4.3 Android Manifest

The Android Manifest is a configuration file that defines essential information about an Android application, including its identifier, permissions and required system capabilities. For GeoMedia, the manifest was manually modified (from the React Native generated) to allow clear-text HTTP traffic, enabling users to configure their own server endpoints and adapt the application to different environments.

A snippet of some permissions and the clear traffic usage (disabled by default since Android 8) is shown in Listing

A snippet of some of configurated permissions and the clear-text traffic option is shown in Listing 5.5. The usesCleartextTraffic attribute was explicitly enabled because Android 9 (API level 28) introduced a default restriction on unencrypted HTTP communication, requiring applications that rely on HTTP connections to opt in explicitly.

```
<manifest xmlns:android="http://schemas.android.com/apk/res/android">
  <uses-permission android:name="android.permission.ACCESS_COARSE_LOCATION"/>
  <uses-permission android:name="android.permission.ACCESS_FINE_LOCATION"/>
  <uses-permission android:name="android.permission.CAMERA"/>
  <uses-permission android:name="android.permission.READ_EXTERNAL_STORAGE"/>
  <uses-permission android:name="android.permission.WRITE_EXTERNAL_STORAGE"/>
  <uses-permission android:name="android.permission.VIBRATE"/>
```

```
  ...

  <application android:usesCleartextTraffic="true"
    android:name=".MainApplication" ...>
    <meta-data android:name="com.google.android.geo.API_KEY" android:value="..."/>
  </application>
</manifest>
```

Listing 5.5: Android Manifest snippet.

Chapter 6

Geomedia Backend

6.1 REST Server

HTTP vs HTTPS, brief notion on SSL/HTTPS security
> wireshark?

6.2 Python FastAPI

6.2.1 NodeJS HTTP Comparison

6.2.2 Swagger and aanalysis

6.3 Cybersecurity

6.3.1 Authentication

6.3.2 SQL Injections

Chapter 7

Data Layer

7.1 DBMS

what it is, what can do

7.1.1 MSSQL

7.1.2 Docker Deployment

Chapter 8

Conclusions

8.1 Consuntivo finale

8.2 Raggiungimento degli obiettivi

8.3 Conoscenze acquisite

8.4 Valutazione personale

“...”

— ....

Acknowledgements

Innanzitutto, vorrei esprimere la mia gratitudine al Prof. Claudio Enrico Palazzi, relatore della mia tesi, per l’aiuto e il sostegno fornitomi durante la stesura del lavoro.

...

...

Padua, September 2026 Alberto Morini

Appendix A

Appendice A

Citazione

Autore della citazione

Acronyms and abbreviations

POI Points of Interest or Point of Interest. 7, 9

Glossary

Metadata in informatica con il termine *Application Programming Interface API* (ing. interfaccia di programmazione di un'applicazione) si indica ogni insieme di procedure disponibili al programmatore, di solito raggruppate a formare un set di strumenti specifici per l'espletamento di un determinato compito all'interno di un certo programma. La finalità è ottenere un'astrazione, di solito tra l'hardware e il programmatore o tra software a basso e quello ad alto livello semplificando così il lavoro di programmazione. 7, 45

Bibliography

Article and papers

[1] Lorenzo Perinello. Alberto Morini Claudio Enrico Palazzi. *GoodIT 2025 - International Conference on Information Technology for Social Good.* 2025. URL: https://goodit2025.org (cit. on p. ii).

[3] Leon Ciechanowski Amin Mahmoudi Dariusz Jemielniak. *Echo Chambers in Online Social Networks: A Systematic Literature Review.* 2024. URL: https://ieeexplore.ieee.org/abstract/document/10388309 (cit. on p. 2).

[6] Aoxia Chen and Pankaj K. Sen. *Advancement in battery technology: A state-of-the-art review.* 2016. URL: https://ieeexplore.ieee.org/document/7731812 (cit. on p. 1).

[9] Ophir Frieder. Eugene Yang David D. Lewis. *Social Network Sites: Definition, History, and Scholarship Open Access.* 2021. URL: https://arxiv.org/abs/2108.12752 (cit. on p. 3).

[10] Giulia Fioravanti et al. *Fear of missing out and social networking sites use and abuse A meta-analysis.* 2021. URL: https://www.sciencedirect.com/science/article/abs/pii/S074756322100162X (cit. on p. 5).

[32] Yuhao Zhu Matthew Halpern and Vijay Janapa Reddi. *Mobile CPU's rise to power: Quantifying the impact of generational mobile CPU design trends on performance, energy, and user satisfaction.* 2016. URL: https://ieeexplore.ieee.org/document/7446054 (cit. on p. 1).

[33] Hany Farid. Matyáš Boháček. *Protecting President Zelenskyy against Deep Fakes.* 2022. URL: https://arxiv.org/abs/2206.12043 (cit. on p. 2).

[34] Zahra Khanbabaloo. Mohammad Hajarian. *Toward Stopping Incel Rebellion: Detecting Incels in Social Media Using Sentiment Analysis.* 2021. URL: https://www.researchgate.net/profile/Mohammad-Hajarian/publication/352100924_Toward_Stopping_Incel_Rebellion_Detecting_Incels_in_Social_Media_Using_Sentiment_Analysis/links/60b8e34992851cb13d73e858/Toward-Stopping-Incel-Rebellion-Detecting-Incels-in-Social-Media-Using-Sentiment-Analysis.pdf (cit. on p. 2).

[41] Andrew Tomk. Ravi Kum Jasmine Novak. *Structure and Evolution of Online Social Networks.* 2006. URL: https://dl.acm.org/doi/abs/10.1145/1150402.1150476 (cit. on p. 2).

[43] Dmitri Rozgonjuk et al. *Fear of Missing Out (FoMO) and social media's impact on daily life and productivity at work: Do WhatsApp, Facebook, Instagram, and Snapchat use disorders mediate that association?* 2020. URL: https://www.sciencedirect.com/science/article/abs/pii/S0306460320306171 (cit. on p. 5).

[45] Roger W. Sinnott. *Virtues of the Haversine.* 1984 (cit. on p. 17).

[55] T. Vincenty. *Direct and inverse solutions of geodesics on the ellipsoid with application of nested equations.* 1975. URL: https://www.tandfonline.com/doi/abs/10.1179/sre.1975.23.176.88 (cit. on p. 17).

Web sites

[2] Anitha Edison Amala Mary. *Deep fake Detection using deep learning techniques A Literature Review.* 2023. URL: https://ieeexplore.ieee.org/abstract/document/10164881/authors#authors (cit. on p. 3).

[4] Shariq Aziz Butt et al. *Remote mobile health monitoring frameworks and mobile applications: Taxonomy, open challenges, motivation, and recommendations.* 2024. URL: https://www.sciencedirect.com/science/article/abs/pii/S0952197624003919 (cit. on p. 1).

[5] Windows Central. *Windows Phone isn't dead, Part VI: App Gap? Microsoft has a platform for that.* 2016. URL: https://www.windowscentral.com/windows-phone-isnt-dead-part-vi-app-gap (cit. on p. 1).

[7] Tech Crunch. *Instagram rolls out a new searchable map to make it easier to discover popular locations.* 2022. URL: https://techcrunch.com/2022/07/19/instagram-new-searchable-map-experience/ (cit. on p. 7).

[8] Demandsage. *Android usage in 2026.* 2026. URL: https://www.demandsage.com/android-statistics/ (cit. on p. 35).

[11] Google. *Google Maps pricing up to 2026.* 2026. URL: https://developers.google.cn/maps/billing-and-pricing/pricing?hl=en (cit. on p. 29).

[12] Waze Mobile Ltd. Google. *Waze official page.* 2006. URL: https://www.waze.com/apps (cit. on p. 1).

[13] Jeffrey Gottfried. *Americans' Social Media Use: YouTube and Facebook are by far the most used online platforms among U.S. adults; TikTok's user base has grown since 2021.* 2024. URL: https://www.pewresearch.org/internet/2024/01/31/americans-social-media-use/ (cit. on p. 2).

[14] Apple Inc. *iOS official page.* 2007. URL: https://www.apple.com/os/ios/ (cit. on p. 1).

[15] Augglin Inc. *Auglinn official page*. 2024. URL: https://www.auglinn.com/ (cit. on p. 10).

[16] Google Inc. *Android Auto official page*. 2015. URL: https://www.android.com/auto/ (cit. on p. 9).

[17] Google Inc. *Android official page*. 2008. URL: https://www.android.com/ (cit. on p. 1).

[18] Google Inc. *Google Lens official page*. 2017. URL: https://lens.google/ (cit. on p. 1).

[19] Google Inc. *Google Maps official page*. 2005. URL: https://www.google.com/maps/about/ (cit. on pp. 1, 9).

[20] Google Inc. *Google shuts failed social network Google*. 2019. URL: https://www.bbc.com/news/technology-47771927 (cit. on p. 5).

[21] Google Inc. *Google Street View*. 2007. URL: https://www.google.com/streetview/ (cit. on p. 9).

[22] Google Inc. *See where your friends are with Google Latitude*. 2009. URL: https://googleblog.blogspot.com/2009/02/see-where-your-friends-are-with-google.html (cit. on p. 5).

[23] Jagat Inc. *Jagat official page*. 2023. URL: https://www.jagat.io/ (cit. on p. 10).

[24] Meta Inc. *Meta Reports Fourth Quarter and Full Year 2023 Results; Initiates Quarterly Dividend*. 2024. URL: https://investor.atmeta.com/investor-news/press-release-details/2024/Meta-Reports-Fourth-Quarter-and-Full-Year-2023-Results-Initiates-Quarterly-Dividend/default.aspx (cit. on p. 2).

[25] Meta Inc. *Scaling the Instagram Explore recommendations system*. 2023. URL: https://engineering.fb.com/2023/08/09/ml-applications/scaling-instagram-explore-recommendations-system/ (cit. on p. 7).

[26] Snap Inc. *Snap Inc. Announces Third Quarter 2025 Financial Results*. 2025. URL: https://investor.snap.com/news/news-details/2025/Snap-Inc--Announces-Third-Quarter-2025-Financial-Results/default.aspx (cit. on p. 5).

[27] Snap Inc. *Snapchat official page*. 2011. URL: https://www.snapchat.com/ (cit. on p. 5).

[28] Marina Niessner. J. Anthony Cookson William Mullins. *Social Media and Finance*. 2024. URL: https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4806692 (cit. on p. 2).

[29] Steven Furnell. Karl van der Schyff Stephen Flowerday. *Duplicitous social media and data surveillance An evaluation of privacy risk*. 2020. URL: https://www.sciencedirect.com/science/article/abs/pii/S0167404820300961 (cit. on p. 2).

[30] Andrei O. J. Kwok and Sharon G. M. Koh. *Deepfake: a social construction of technology perspective.* 2020. URL: https://www.tandfonline.com/doi/abs/10.1080/13683500.2020.1738357 (cit. on p. 2).

[31] Peter James Marcin Straczkiewicz and Jukka-Pekka Onnela. *A systematic review of smartphone-based human activity recognition methods for health research.* 2021. URL: https://www.nature.com/articles/s41746-021-00514-4 (cit. on p. 1).

[35] ABC News. *Snapchat's new Snap Map feature raises privacy concerns.* 2017. URL: https://abcnews.com/Lifestyle/snapchats-snap-map-feature-raises-privacy-concerns/story?id=48271889 (cit. on p. 5).

[36] Neliia Blynova Oksana Kyrylova and Viktoriia Pavlenko. *Mobile internet usage worldwide - statistics and facts.* 2023. URL: https://www.statista.com/topics/779/mobile-internet/.

[37] Neliia Blynova Oksana Kyrylova and Viktoriia Pavlenko. *The perspectives for mobile application use in media education.* 2023. URL: https://www.tandfonline.com/doi/full/10.1080/10494820.2023.2186897 (cit. on p. 1).

[38] S.A. OutSystems- Software em Rede. *IonicFramework official page.* 2013. URL: https://ionicframework.com/ (cit. on p. 26).

[39] A. Perrin. *Social Media Usage: 2005–2015 – 65% of adults now use social networking sites, a nearly tenfold jump in the past decade.* 2015. URL: https://www.secretintelligenceservice.org/wp-content/uploads/2016/02/PI_2015-10-08_Social-Networking-Usage-2005-2015_FINAL.pdf (cit. on p. 2).

[40] Amanda Raffoul et al. *Social media platforms generate billions of dollars in revenue from U.S. youth Findings from a simulated revenue model.* 2023. URL: https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0295337 (cit. on p. 2).

[42] Reuters. *Meta CEO Zuckerberg says Instagram has grown to 3 billion monthly active users.* 2025. URL: https://www.reuters.com/business/meta-ceo-zuckerberg-says-instagram-has-grown-3-billion-monthly-active-users-2025-09-24/ (cit. on p. 6).

[44] Apple Inc. Shazam Entertainment Limited. *Shazam official page.* 2002. URL: https://www.shazam.com/ (cit. on p. 1).

[46] Meta Open Source. *React Native directory for external packages.* 2015. URL: https://reactnative.directory/ (cit. on p. 27).

[47] Meta Open Source. *React Native official page.* 2015. URL: https://reactnative.dev/ (cit. on p. 25).

[48] Statista. *Device usage of Facebook users worldwide as of January 2022.* 2022. URL: https://www.statista.com/statistics/377808/distribution-of-facebook-users-by-device/ (cit. on p. 1).

[49] Statista. *Global smartphone sales to end users from 2007 to 2023*. 2023. URL: https://www.statista.com/statistics/263437/global-smartphone-sales-to-end-users-since-2007/ (cit. on p. 1).

[50] Statista. *M Number of monthly active Instagram users from January 2013 to December 2021 .* 2021. URL: https://www.statista.com/statistics/253577/number-of-monthly-active-instagram-users/ (cit. on p. 6).

[51] Statista. *Most popular global mobile messenger apps as of October 2025, based on number of monthly active users*. 2025. URL: https://www.statista.com/statistics/258749/most-popular-global-mobile-messenger-apps/ (cit. on p. 1).

[52] Social Media Today. *Instagram Stories is Now Being Used by 500 Million People Daily*. 2019. URL: https://www.socialmediatoday.com/news/instagram-stories-is-now-being-used-by-500-million-people-daily/547270/ (cit. on p. 7).

[53] Eastern District of New York. U.S. Attorney's Office. *Two Individuals Arrested for Publishing AI Deepfake Pornography In Violation Of TAKE IT DOWN Act*. 2026. URL: https://www.justice.gov/usao-edny/pr/two-individuals-arrested-publishing-ai-deepfake-pornography-violation-take-it-down-act (cit. on p. 2).

[54] International Telecommunication Union. *International Telecommunication Union. 2023. Facts and Figures 2023 – Mobile Phone Ownership*. 2023. URL: https://www.itu.int/itu-d/reports/statistics/2023/10/10/ff23-mobile-phone-ownership/ (cit. on p. 1).

[56] Computer World. *Windows Phone still has serious app gap despite Microsoft's $7.2 billion Nokia buyout*. 2013. URL: https://www.computerworld.com/article/1500308/windows-phone-still-has-serious-app-gap-despite-microsoft-s-7-2-billion-nokia-buyo.html (cit. on p. 1).